

Permanental Dominance in Dimension Four

A Direct Geometric Proof

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Abstract

For a subgroup $H \leq \mathfrak{S}_4$ and an irreducible complex character χ of H , let

$$d_\chi^H(A) = \sum_{\sigma \in H} \chi(\sigma) \prod_{i=1}^4 A_{i, \sigma(i)}$$

be the associated generalized matrix function. We prove that every Hermitian positive-semidefinite matrix $A \in M_4(\mathbb{C})$ satisfies

$$\frac{1}{\chi(1)} d_\chi^H(A) \leq \text{per } A.$$

Both sides are real. The proof first reduces to correlation matrices and then treats the thirty-seven irreducible-character rows belonging to the eleven conjugacy types of subgroups of $\mathfrak{S}_4$. All but two rows follow from elementary principal-minor, moment, and block-contraction inequalities. The two remaining rows are the non-real one-dimensional characters of A_4 . For them the difference from the permanent becomes an affine Hermitian quadratic form. A residual-vector normalization turns this form into a polynomial on the cone of 3×3 Gram matrices. We prove its nonnegativity by a rank-one sum of squares, a positive-definite two-dimensional Hessian, and exact rational Gram certificates. The argument also yields a positive-semidefinite adjugate certificate for the exceptional A_4 rows.

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1 Introduction

The normalized permanental-dominance problem compares two ways of summing permutation monomials of a positive-semidefinite matrix. The permanent uses every permutation with coefficient one. A generalized matrix function uses only a subgroup and weights its elements by an irreducible character. The claim is that, after division by the character degree, none of these weighted sums exceeds the permanent.

In dimension four the statement is finite but not automatic. There are eleven conjugacy types of subgroups of $\mathfrak{S}_4$ and thirty-seven irreducible character rows. Thirty-five rows are controlled by familiar inequalities for correlation matrices. The genuine obstruction is concentrated in the two conjugate non-real one-dimensional characters of A_4 . Thus the group theory is essential for completeness, but the mathematical core of the proof is a single geometric inequality.

The organizing idea is to write the exceptional gap as

$$c_0(u, v, w) + q^* K(u, v, w) q.$$

The block positivity constraint says that $B(u, v, w) - qq^*$ is a Gram matrix. After dividing out the nonzero coordinates of q , this residual becomes the Gram matrix R of three vectors, and the exceptional gap factors as

$$|q_1 q_2 q_3|^2 \mathcal{P}(R).$$

The problem is therefore moved from a character sum on four indices to the nonnegativity of one explicit polynomial on the cone of 3×3 Gram matrices.

The complete proof may be read through the following chain:

$$\begin{aligned} A \succeq 0 &\longrightarrow \text{a correlation matrix} \longrightarrow \text{thirty-seven representative character rows} \\ &\longrightarrow \text{thirty-five routine rows plus two non-real } A_4 \text{ rows} \\ &\longrightarrow c_0 + q^* K q \longrightarrow |q_1 q_2 q_3|^2 \mathcal{P}(R) \\ &\longrightarrow \begin{cases} \text{a rank-one sum of squares,} \\ \text{a positive-definite Hessian/minimum certificate,} \\ \text{closure on the positive-semidefinite boundary.} \end{cases} \end{aligned}$$

Section 2 fixes the normalization and removes the diagonal parameters. Section 3 proves that the finite list of rows is complete, and Section 4 eliminates the thirty-five routine rows. Sections 5 and 6 contain the geometric argument. The proof is assembled in Section 7; exact coefficient certificates and the Lean correspondence are kept in the appendices so that they do not interrupt the main line of reasoning.

2 Statement and general reductions

Let $n \geq 1$, let $H \leq \mathfrak{S}_n$, and let $\chi : H \rightarrow \mathbb{C}$ be a character. For $A = (A_{ij}) \in M_n(\mathbb{C})$ put

$$d_\chi^H(A) := \sum_{\sigma \in H} \chi(\sigma) \prod_{i=1}^n A_{i, \sigma(i)}, \quad \text{per } A := \sum_{\sigma \in \mathfrak{S}_n} \prod_{i=1}^n A_{i, \sigma(i)}. \quad (1)$$

The normalization in the permanental-dominance conjecture is $d_\chi^H / \chi(1)$. Notice that no complex conjugate is placed on $\chi(\sigma)$ in (1); this convention agrees with all formulae below.

Lemma 2.1 (Reality). *If $A = A^*$ and χ is the character of a finite-dimensional complex representation of H , then $d_\chi^H(A)$ and $\text{per } A$ are real.*

Proof. Every finite-group representation may be made unitary, and hence $\overline{\chi(\sigma)} = \chi(\sigma^{-1})$. Hermitian symmetry gives

$$\overline{\prod_i A_{i,\sigma(i)}} = \prod_i A_{\sigma(i),i} = \prod_j A_{j,\sigma^{-1}(j)}.$$

Conjugating the sum and replacing σ by σ^{-1} proves the first claim. The second is the special case $H = \mathfrak{S}_n$, $\chi = 1$. $\square$

Theorem 2.2 (Permanental dominance for $n = 4$). *Let $H \leq \mathfrak{S}_4$, let V be an irreducible finite-dimensional complex representation of H , and let χ be its character. For every $A \in M_4(\mathbb{C})$ with $A \succeq 0$,*

$$\frac{d_\chi^H(A)}{\dim V} \leq \text{per } A. \quad (2)$$

The scope of Theorem 2.2 is exactly dimension four. It includes every subgroup of $\mathfrak{S}_4$ and every irreducible complex representation of that subgroup, but it does not assert the conjecture in arbitrary dimension.

2.1 Reduction to correlation matrices

The normalization step is elementary but important because it removes all diagonal parameters without an invertibility hypothesis.

Lemma 2.3 (Diagonal congruence). *Let $s_1, \dots, s_n \in \mathbb{R}$ and $D = \text{diag}(s_1, \dots, s_n)$. Then*

$$d_\chi^H(DAD) = \left(\prod_i s_i \right)^2 d_\chi^H(A), \quad \text{per}(DAD) = \left(\prod_i s_i \right)^2 \text{per } A.$$

Proof. For every permutation σ ,

$$\prod_i (DAD)_{i,\sigma(i)} = \prod_i (s_i A_{i,\sigma(i)} s_{\sigma(i)}) = \left(\prod_i s_i \right)^2 \prod_i A_{i,\sigma(i)}.$$

Summation proves both identities. $\square$

Lemma 2.4 (Correlation reduction). *It is enough to prove Theorem 2.2 for positive-semidefinite matrices whose diagonal entries are all one.*

Proof. Suppose first that $A_{ii} > 0$ for all i and put $s_i = \sqrt{A_{ii}}$. Since a Hermitian matrix has real diagonal, the matrix

$$C = \text{diag}(s_i^{-1}) A \text{diag}(s_i^{-1})$$

is positive semidefinite and has diagonal one. Moreover $A = \text{diag}(s_i) C \text{diag}(s_i)$, so Lemma 2.3 transfers the desired inequality from C to A .

If $A_{ii} = 0$ for some i , positivity of the 2×2 principal minors implies $A_{ij} = A_{ji} = 0$ for every j . Each permutation monomial then contains a zero entry in row i , so both sides of (2) vanish. $\square$

We therefore write a general 4×4 correlation matrix as

$$A(a, b, c, d, e, f) = \begin{pmatrix} 1 & a & b & c \\ \bar{a} & 1 & d & e \\ \bar{b} & \bar{d} & 1 & f \\ \bar{c} & \bar{e} & \bar{f} & 1 \end{pmatrix} \succeq 0. \quad (3)$$

3 Finite character reduction

3.1 Subgroups and characters

Up to conjugacy, the subgroups of $\mathfrak{S}_4$ have the following eleven types. Here $r = (1234)$ and $s = (13)$ in the dihedral row.

type	representative	abstract group	order	irreducible rows
1	$\{1\}$	1	1	1
C_2^t	$\langle(12)\rangle$	C_2	2	2
C_2^d	$\langle(12)(34)\rangle$	C_2	2	2
C_3	$\langle(123)\rangle$	C_3	3	3
C_4	$\langle(1234)\rangle$	C_4	4	4
V_4^n	$\{1, (12)(34), (13)(24), (14)(23)\}$	C_2^2	4	4
V_4^d	$\langle(12), (34)\rangle$	C_2^2	4	4
S_3	$\text{Stab}(4)$	S_3	6	3
D_8	$\langle r, s \rangle$	D_8	8	5
A_4	A_4	A_4	12	4
S_4	S_4	S_4	24	5

The row counts sum to 37.

Lemma 3.1 (Subgroup classification). *Every subgroup of $\mathfrak{S}_4$ is conjugate to exactly one of the eleven types in the preceding table.*

Proof. We recall the elementary classification. An intransitive subgroup preserves an orbit partition of 4. The partitions $1 + 1 + 1 + 1$, $2 + 1 + 1$, $2 + 2$ and $3 + 1$ give respectively 1, the two possible embedded copies of C_2 , V_4^d and subgroups of the displayed S_3 . This yields 1, C_2^t , C_2^d , C_3 , V_4^d , S_3 .

Now suppose the action is transitive. Orbit-stabilizer shows that the order is divisible by 4 and divides 24. A Sylow analysis gives the following possibilities. A cyclic Sylow 2-subgroup generated by a four-cycle gives C_4 ; a non-cyclic subgroup of order 4 is the normal double-transposition group V_4^n ; a subgroup of order 8 is a Sylow 2-subgroup and is conjugate to $\langle(1234), (13)\rangle$; order 12 gives the unique index-two subgroup A_4 ; and order 24 gives S_4 . These groups have different order, orbit structure, or involution type, so the listed types are not conjugate to one another. $\square$

The character data used below are recorded in Appendix A. Orthogonality of those rows and the identity $\sum_{\chi} \chi(1)^2 = |H|$ prove that no irreducible character is missing. For the two exceptional A_4 rows we fix

$$\omega = -\frac{1}{2} - \frac{\sqrt{3}}{2}i, \quad \omega^2 + \omega + 1 = 0. \quad (4)$$

3.2 Conjugation invariance

Lemma 3.2 (Transport). *Let $g \in \mathfrak{S}_4$, $K = g^{-1}Hg$, and let $\chi^g(g^{-1}\sigma g) = \chi(\sigma)$. If $A_{ij}^g = A_{g(i),g(j)}$, then*

$$d_{\chi^g}^K(A^g) = d_{\chi}^H(A), \quad \text{per } A^g = \text{per } A, \quad A \succeq 0 \implies A^g \succeq 0.$$

Proof. Simultaneous permutation of rows and columns preserves positivity and the permanent. For a permutation monomial, changing i to $g(i)$ gives

$$\prod_i A_{i,(g^{-1}\sigma g)(i)}^g = \prod_j A_{j,\sigma(j)}.$$

Summing proves the assertion. $\square$

Thus it suffices to prove dominance for the thirty-seven rows of the eleven fixed representatives.

4 Routine character rows

4.1 Four cycle aggregates

For (3) define

$$T = |a|^2 + |b|^2 + |c|^2 + |d|^2 + |e|^2 + |f|^2, \quad (5)$$

$$D = |a|^2|f|^2 + |b|^2|e|^2 + |c|^2|d|^2, \quad (6)$$

$$C = 2 \operatorname{Re}(adb + ae\bar{c} + bf\bar{c} + df\bar{e}), \quad (7)$$

$$F = 2 \operatorname{Re}(adf\bar{c} + aef\bar{b} + bde\bar{c}). \quad (8)$$

Enumerating the cycle types of $\mathfrak{S}_4$ gives

$$\operatorname{per} A = 1 + T + D + C + F. \quad (9)$$

The five normalized S_4 rows in the order (50) are

$$\begin{aligned} p_{[4]} &= 1 + T + D + C + F, \\ p_{[1^4]} &= 1 - T + D + C - F, \\ p_{[31]} &= 1 + \frac{1}{3}T - \frac{1}{3}D - \frac{1}{3}F, \\ p_{[211]} &= 1 - \frac{1}{3}T - \frac{1}{3}D + \frac{1}{3}F, \\ p_{[22]} &= 1 + D - \frac{1}{2}C. \end{aligned} \quad (10)$$

Lemma 4.1 (Scalar correlation inequalities). *If A in (3) is positive semidefinite, then*

$$0 \leq T, \quad 0 \leq D, \quad D \leq \frac{T^2}{4}, \quad -2D \leq F \leq 2D \leq T, \quad (11)$$

$$\frac{T^2}{2} \leq T + \frac{3}{2}C, \quad 1 \leq \operatorname{per} A. \quad (12)$$

Every 3×3 principal permanent is at most $\operatorname{per} A$.

Proof. The inequalities $|a|^2, \dots, |f|^2 \leq 1$ follow from the 2×2 principal minors. The first four estimates in (11) are then AM-GM and $2|\operatorname{Re}(xy)| \leq |x|^2 + |y|^2$, paired over opposite edges.

For the moment inequality, let $\lambda_1, \dots, \lambda_4 \geq 0$ be the eigenvalues of A . Direct trace expansion gives

$$\sum \lambda_i = 4, \quad \sum \lambda_i^2 = 4 + 2T, \quad \sum \lambda_i^3 = 4 + 6T + 3C.$$

The exact identity

$$\left(\sum_i \lambda_i \right) \left(\sum_i \lambda_i^3 \right) - \left(\sum_i \lambda_i^2 \right)^2 = \sum_{i < j} \lambda_i \lambda_j (\lambda_i - \lambda_j)^2$$

implies (12)'s first inequality. Combining it with $D \leq T^2/4$ and $F \geq -2D$ gives $1 \leq 1 + T + D + C + F$.

Finally, the permanent version of the $3 + 1$ Fischer contraction can be verified directly from (9): after subtracting a principal 3×3 permanent, the remaining expression is the sum of the corresponding 3×3 determinant inequality and squared edge terms. Applying this to each of the four principal triples proves the last claim. $\square$

We also need two symmetric $2 + 2$ contractions. Put

$$R_+ := 1 + |b|^2 + |e|^2 + D + 2 \operatorname{Re}(adf\bar{c}), \quad (13)$$

$$R_\wedge := (1 - |b|^2)(1 - |e|^2) + |af - cd|^2. \quad (14)$$

Lemma 4.2 (Block contractions). *For a positive-semidefinite correlation matrix,*

$$R_+ \leq \text{per } A, \quad R_\wedge \leq \text{per } A, \quad 1 + |a|^2|f|^2 + |c|^2|d|^2 \leq \text{per } A. \quad (15)$$

Proof. For the split $\{1, 3\} \mid \{2, 4\}$, take Gram vectors x_1, x_2, x_3, x_4 for A . Orthogonally remove the component of x_3 along x_1 and apply Cauchy–Schwarz to the resulting two pairs. Expansion gives the first inequality. Applying exterior Cauchy–Schwarz to $x_1 \wedge x_3$ and $x_2 \wedge x_4$ gives

$$|af - c\bar{d}|^2 \leq (1 - |b|^2)(1 - |e|^2),$$

and the same block expansion gives the second inequality. Relabelling the split twice yields $(1 + |a|^2)(1 + |f|^2) \leq \text{per } A$ and $(1 + |c|^2)(1 + |d|^2) \leq \text{per } A$; comparison of the two opposite-edge products gives the third inequality. $\square$

4.2 Explicit dispatch of the routine character rows

The following proposition records the finite analytic part of the proof. Its table is included to make clear that the only rows not covered are the two non-real A_4 rows.

Proposition 4.3 (Thirty-five routine rows). *For every representative subgroup in Lemma 3.1, all normalized irreducible rows other than χ_ω and $\bar{\chi}_\omega$ of A_4 are bounded above by $\text{per } A$.*

Proof. The cyclic, Klein-four, and S_3 expansions are as follows. In the C_3 and S_3 rows put $z = adb$.

group	normalized row values
1	1
C_2^t	$1 \pm a ^2$
C_2^d	$1 \pm a ^2 f ^2$
C_3	$1 + 2 \operatorname{Re} z, \quad 1 - \operatorname{Re} z \pm \sqrt{3} \operatorname{Im} z$
V_4^n	$1 \pm r \pm s \pm t$ with the four character sign patterns, $r = a ^2 f ^2, s = b ^2 e ^2, t = c ^2 d ^2$
V_4^d	$(1 \pm a ^2)(1 \pm f ^2)$
S_3	$1 + a ^2 + b ^2 + d ^2 + 2 \operatorname{Re} z,$ $1 - a ^2 - b ^2 - d ^2 + 2 \operatorname{Re} z, \quad 1 - \operatorname{Re} z$

These expressions follow by listing the elements of the corresponding subgroup. The 3×3 principal determinant inequalities, the principal permanent contraction in Lemma 4.1, and $|a|^2, \dots, |f|^2 \leq 1$ prove their bounds. For example, the three C_3 rows are precisely the normalized one-dimensional rows of the principal 3×3 correlation block, while the three S_3 rows are its trivial, sign, and standard rows.

For C_4 , put

$$r = |b|^2|e|^2, \quad z = adf\bar{c}.$$

Its four rows are

$$1 + r + 2 \operatorname{Re} z, \quad 1 - r - 2 \operatorname{Im} z, \quad 1 + r - 2 \operatorname{Re} z, \quad 1 - r + 2 \operatorname{Im} z. \quad (16)$$

The estimates

$$2|\operatorname{Re} z|, 2|\operatorname{Im} z| \leq |a|^2|f|^2 + |c|^2|d|^2$$

together with Lemma 4.2 prove all four bounds.

For D_8 , write $p = |b|^2, q = |e|^2$. The five rows are

$$R_+, \quad (1 + p)(1 + q) - |af + c\bar{d}|^2, \\ (1 - p)(1 - q) - |af - c\bar{d}|^2, \quad R_\wedge, \quad 1 - pq. \quad (17)$$

The first and fourth are (15); the second is no larger than R_+ by direct expansion; the third and fifth are at most 1, hence at most per A by Lemma 4.1.

For S_4 , substitute (11)–(12) into the five expressions (10). The sign row uses $T + F \geq 0$; the [31] and [22] rows use the moment inequality; and the [211] row is at most 1 because $F \leq T$ and $D \geq 0$.

Finally, the trivial A_4 row is the average of the trivial and sign S_4 rows, while its three-dimensional row is the average of the [31] and [211] rows. This proves the asserted thirty-five rows. $\square$

5 The exceptional A_4 geometry

5.1 The exceptional character gap

The preceding section deliberately has an eliminative role: it reduces the entire theorem to two conjugate character rows. We now leave the finite table calculation behind and treat their common geometric content.

Let

$$B(u, v, w) = \begin{pmatrix} 1 & u & v \\ \bar{u} & 1 & w \\ \bar{v} & \bar{w} & 1 \end{pmatrix}. \quad (18)$$

Define

$$c_0(u, v, w) = |u|^2 + |v|^2 + |w|^2 + 2 \operatorname{Re}((1 - \omega)uw\bar{v}) \quad (19)$$

and the Hermitian matrix

$$K(u, v, w) = \begin{pmatrix} 1 & (1 - \omega^2)u + v\bar{w} & uw + (1 - \omega)v \\ (1 - \omega)\bar{u} + w\bar{v} & 1 & (1 - \omega^2)w + v\bar{u} \\ \bar{u}\bar{w} + (1 - \omega^2)\bar{v} & (1 - \omega)\bar{w} + u\bar{v} & 1 \end{pmatrix}. \quad (20)$$

Lemma 5.1 (Character expansion). *For the correlation matrix (3),*

$$\operatorname{per} A - d_{\chi_\omega}^{A_4}(A) = c_0(a, b, d) + \begin{pmatrix} \bar{c} & \bar{e} & \bar{f} \end{pmatrix} K(a, b, d) \begin{pmatrix} c \\ e \\ f \end{pmatrix}. \quad (21)$$

For $\bar{\chi}_\omega$, the same formula holds after an odd simultaneous relabeling of the four indices.

Proof. List the identity, three double transpositions, and the two classes of four three-cycles in A_4 , insert their values from (51), and pair inverse permutations. Using $\omega^2 = -1 - \omega$ gives exactly (19)–(20). An odd permutation interchanges the two three-cycle classes and therefore exchanges χ_ω with its conjugate. $\square$

The remaining task is consequently the following four-vector inequality.

Theorem 5.2 (Four-vector inequality). *Suppose*

$$\mathcal{A} = \begin{pmatrix} B(u, v, w) & q \\ q^* & 1 \end{pmatrix} \succeq 0, \quad q \in \mathbb{C}^3. \quad (22)$$

Then

$$c_0(u, v, w) + q^* K(u, v, w) q \geq 0. \quad (23)$$

The proof is developed in the residual normalization below and in Section 6.

5.2 Residual-vector normalization

There is a useful geometric meaning behind the next change of variables. Realize the block matrix (22) as the Gram matrix of four unit vectors. The vector corresponding to the last row and column determines a common one-dimensional direction. Subtracting the projections of the first three vectors onto that direction leaves three residual vectors, whose Gram matrix is $B - qq^*$. Dividing the residual vectors by the coordinates of the removed projections does not change positivity; it merely produces the scale-free matrix R below. This is the step that turns a four-vector completion problem into a problem on the cone of three-vector Gram matrices.

Assume first that $q_1 q_2 q_3 \neq 0$. The Schur complement of the final diagonal entry in (22) is $B - qq^* \succeq 0$. Congruence by $\text{diag}(\bar{q}_1^{-1}, \bar{q}_2^{-1}, \bar{q}_3^{-1})$ gives the residual Gram matrix

$$R = \begin{pmatrix} n_1 & h_{12} & h_{13} \\ \bar{h}_{12} & n_2 & h_{23} \\ \bar{h}_{13} & \bar{h}_{23} & n_3 \end{pmatrix} \succeq 0, \quad (24)$$

where

$$n_i = \frac{1 - |q_i|^2}{|q_i|^2}, \quad h_{12} = \frac{u}{q_1 \bar{q}_2} - 1, \quad h_{13} = \frac{v}{q_1 \bar{q}_3} - 1, \quad h_{23} = \frac{w}{q_2 \bar{q}_3} - 1. \quad (25)$$

Thus R is the Gram matrix of three residual vectors w_1, w_2, w_3 with $n_i = \|w_i\|^2$ and $h_{ij} = \langle w_i, w_j \rangle$.

We now define the normalized geometric polynomial. Put

$$\begin{aligned} E &= n_1 n_2 + |h_{12}|^2 + n_1 n_3 + |h_{13}|^2 + n_2 n_3 + |h_{23}|^2, \\ x_{12,13} &= n_1 h_{23} + h_{13} \bar{h}_{12}, \\ x_{12,23} &= h_{12} h_{23} + n_2 h_{13}, \\ x_{13,23} &= n_3 h_{12} + h_{13} \bar{h}_{23}, \\ S &= E + 2 \operatorname{Re}(x_{12,13} + x_{12,23} + x_{13,23}), \\ S_\omega &= E + 2 \operatorname{Re}(\omega^2 x_{12,13} + \omega x_{12,23} + \omega^2 x_{13,23}), \\ \tau &= h_{13} \bar{h}_{12} \bar{h}_{23}, \\ \Gamma &= n_3 |h_{12}|^2 + n_2 |h_{13}|^2 + n_1 |h_{23}|^2 + 3 \operatorname{Re} \tau + \sqrt{3} \operatorname{Im} \tau. \end{aligned} \quad (26)$$

Finally set

$$\mathcal{P}(R) = 24 + 6(n_1 + n_2 + n_3 + 2 \operatorname{Re}(h_{12} + h_{13} + h_{23})) + 2S - S_\omega + \Gamma. \quad (27)$$

Lemma 5.3 (Exact normalization identity). *With (25),*

$$c_0(u, v, w) + q^* K(u, v, w) q = |q_1 q_2 q_3|^2 \mathcal{P}(R). \quad (28)$$

Proof. Equations (25) give

$$u = q_1 \bar{q}_2 (1 + h_{12}), \quad v = q_1 \bar{q}_3 (1 + h_{13}), \quad w = q_2 \bar{q}_3 (1 + h_{23}),$$

and $|q_i|^2(1 + n_i) = 1$. Substitute these identities in (19) and (20), use $\omega^2 + \omega + 1 = 0$, and collect the coefficients of the three normalization equations. The diagonal quadratic terms give E , the three mixed symmetric-tensor terms give S and S_ω , and the cubic terms give Γ . The result is (28). $\square$

Thus Theorem 5.2 follows from the next result.

Theorem 5.4 (Geometric polynomial). *For every Hermitian matrix R of the form (24),*

$$R \succeq 0 \implies \mathcal{P}(R) \geq 0.$$

6 Positivity of the geometric polynomial

The proof uses the geometry of the Gram cone in three stages. On the rank-one locus the three residual vectors are collinear and the polynomial has a direct sum-of-squares form. In the positive-definite region, two residual vectors are used as coordinates and the third is minimized through an explicit two-dimensional Hermitian Hessian. Finally, every singular Gram matrix is obtained as a limit of positive-definite ones. The rank-one calculation is also the algebraic base coefficient in the interior minimum certificate, so it is both explanatory and logically active.

6.1 The rank-one boundary

Suppose first that R has rank one. Write

$$n_1 = |x|^2, \quad n_2 = |y|^2, \quad n_3 = |z|^2, \quad h_{12} = \bar{x}y, \quad h_{13} = \bar{x}z, \quad h_{23} = \bar{y}z.$$

Define

$$R_0(x, y, z) = 4 + |x + y + z|^2 + |xy + xz + yz|^2 - (|xy|^2 + |xz|^2 + |yz|^2) + |xyz|^2, \quad (29)$$

$$\Phi(x, y, z) = xy + \omega xz + \omega^2 yz. \quad (30)$$

Lemma 6.1 (Rank-one sum of squares). *For all $x, y, z \in \mathbb{C}$,*

$$\mathcal{P}(R) = 2(3R_0(x, y, z) + |\Phi(x, y, z)|^2) \geq 0. \quad (31)$$

Proof. Regard R_0 as a quadratic in z . Put

$$K_0 = |1 + \bar{x}y|^2, \quad L_0 = \overline{x+y} + \bar{x}y(x+y), \quad C_0 = 4 + |x+y|^2.$$

Direct expansion gives

$$R_0 = K_0|z|^2 + 2\operatorname{Re}(L_0z) + C_0$$

and the exact discriminant identity

$$C_0K_0 - |L_0|^2 = 4(1 + \operatorname{Re}(\bar{x}y))^2 \geq 0.$$

Completing the square proves $R_0 \geq 0$. A second direct expansion using $1 + \omega + \omega^2 = 0$ gives the equality in (31); its right-hand side is nonnegative. $\square$

6.2 The positive-definite region

It remains to treat a positive-definite residual Gram matrix. Write

$$a = n_1 > 0, \quad b = n_2, \quad c = n_3, \quad h_{12} = g = r - is, \quad p = h_{13}, \quad q = h_{23}, \quad (32)$$

and put

$$G = \begin{pmatrix} a & g \\ \bar{g} & b \end{pmatrix}, \quad \delta = \det G = ab - r^2 - s^2 > 0, \quad h = \begin{pmatrix} p \\ q \end{pmatrix}. \quad (33)$$

There is a unique $\zeta \in \mathbb{C}^2$ such that $h = G\zeta$. Positivity of R tested on $(-\zeta, 1)$ gives the Schur slack

$$\zeta^* G \zeta \leq c. \quad (34)$$

Set

$$\lambda = 6 + a + b + r^2 + s^2 + 5r - \sqrt{3}s, \quad (35)$$

$$P_0 = 24 + 6(a + b + 2r) + ab + r^2 + s^2, \quad (36)$$

$$\ell = \begin{pmatrix} 6 + (2 - \omega)g + (2 - \omega^2)b \\ 6 + (2 - \omega)a + (2 - \omega^2)\bar{g} \end{pmatrix}, \quad (37)$$

$$M = \begin{pmatrix} 1 + b & 2 - \omega + (1 - \omega)g \\ 2 - \omega^2 + (1 - \omega^2)\bar{g} & 1 + a \end{pmatrix}, \quad (38)$$

$$T = \lambda I_2 + MG. \quad (39)$$

Lemma 6.2 (Quadratic decomposition). *With the preceding notation,*

$$\begin{aligned} \mathcal{P}(R) &= P_0 + 2\operatorname{Re}(\ell^*h) + \lambda c + \operatorname{Re}(h^*Mh) \\ &= Q(\zeta) + \lambda(c - \zeta^*G\zeta), \end{aligned} \quad (40)$$

where

$$Q(\zeta) = P_0 + 2\operatorname{Re}((G\ell)^*\zeta) + \zeta^*(GT)\zeta. \quad (41)$$

Moreover GT is Hermitian.

Proof. Substitution of (32) in (27) gives the first identity. Replacing h by $G\zeta$, using $G = G^*$ and $T = \lambda I + MG$, gives the second. Finally $GT = \lambda G + GMG$ is Hermitian because G and M are Hermitian. $\square$

Lemma 6.3 (Positivity of the scalar part). *Under $a, b \geq 0$ and $r^2 + s^2 \leq ab$, one has $\lambda > 0$.*

Proof. Let $\rho = \sqrt{r^2 + s^2}$. Cauchy–Schwarz and AM–GM give

$$5r - \sqrt{3}s \geq -2\sqrt{7}\rho, \quad a + b \geq 2\rho.$$

Consequently

$$\lambda \geq (\rho - (\sqrt{7} - 1))^2 + 2(\sqrt{7} - 1) > 0. \quad \square$$

The remaining algebra is organized by the Gram defect δ . Define

$$N = \operatorname{Re}(P_0 \det T - \ell^*G \operatorname{adj}(T)\ell). \quad (42)$$

Lemma 6.4 (Hessian and minimum certificates). *For $a > 0$ and $\delta \geq 0$,*

$$a^3 \operatorname{Re} \det T = D_0 + \delta D_1 + \delta^2 D_2, \quad (43)$$

where

$$\begin{aligned} D_0 &= a^3 \lambda_0 H_0 > 0, \\ D_1 &= a((4 + 11a)(r^2 + s^2) + a(20 + a)r - \sqrt{3}a(4 + 3a)s + a(18 + 10a + 3a^2)) > 0, \end{aligned} \quad (44)$$

$$D_2 = a(a + 1)(a + 2) > 0. \quad (45)$$

Here λ_0 denotes (35) with $b = (r^2 + s^2)/a$, and

$$H_0 = 2\lambda_0 - 6 + 4(r^2 + s^2) > 0.$$

Furthermore

$$a^3 N = N_0 + \delta N_1 + \delta^2 N_2 + \delta^3 N_3, \quad (46)$$

with $N_0 \geq 0$ and $N_1, N_2, N_3 > 0$.

Proof. The determinant identity follows by expanding (39) after $b = (r^2 + s^2 + \delta)/a$. Positivity of H_0 follows from Lemma 6.3 and the sharper estimate obtained from the same two inequalities. For D_1 , complete squares in (r, s) ; the residual is

$$4a(72 + 126a + 94a^2 + 26a^3) > 0.$$

The assertion for D_2 is immediate.

The expansion (46) is obtained in the same way from (42). On the boundary $\delta = 0$, the first two residual vectors are collinear. Lemma 6.1, viewed as a quadratic in the third scalar coordinate, implies that its discriminant is nonpositive. In the notation above this gives

$$N_0 = a^3 \lambda_0 (P_0 H_0 - |L'_0|^2) \geq 0$$

for the corresponding line coefficient L'_0 .

The remaining coefficients are explicit. First,

$$N_3 = (a + 2)(a^2 + 6) > 0.$$

Next

$$\begin{aligned} N_2 = & 36(r^2 + s^2) + a(120 + 84r - 36\sqrt{3}s + 36(r^2 + s^2)) \\ & + a^2(60 + 24r - 24\sqrt{3}s + 16(r^2 + s^2)) \\ & + a^3(16 + 2(r^2 + s^2)) + 2a^4. \end{aligned}$$

Completing squares gives the strictly positive residual

$$16a(a^6 + 16a^5 + 112a^4 + 318a^3 + 588a^2 + 936a + 1080).$$

For N_1 , put $X = r$, $Y = \sqrt{3}s$ and $\nu = X^2 + Y^2/3$. Then

$$N_1 = 36\nu^2 + aB_1 + a^2B_2 + a^3B_3 + a^4B_4 + 6a^5, \quad (47)$$

where

$$\begin{aligned} B_1 &= \nu(240 + 168X - 72Y + 54\nu), \\ B_2 &= 432 + 624X - 240Y + 492X^2 + 108Y^2 - 168XY \\ &\quad + (156X - 84Y)\nu + 38\nu^2, \\ B_3 &= 168 + 72X - 168Y + 54X^2 + 46Y^2 - 84XY \\ &\quad + (20X - 24Y)\nu + 6\nu^2, \\ B_4 &= 60 + 24X - 24Y + 14\nu. \end{aligned}$$

Appendix B gives exact certificates for $B_1 \geq 0$, $B_2 > 0$, $B_4 > 0$, and $B_2 + aB_3 + a^2B_4 > 0$. Hence (47) is strictly positive. $\square$

Proposition 6.5 (Positive-definite residual Gram matrices). *If $R \succ 0$, then $\mathcal{P}(R) > 0$.*

Proof. Equations (43) and (44) show $\text{Re det } T > 0$. The leading entry of GT equals a positive rank-one boundary value plus $\delta(a + 1)$, hence it is positive. Since GT is Hermitian and

$$\det(GT) = \delta \det T > 0,$$

the 2×2 matrix GT is positive definite.

The critical point of (41) is

$$\zeta_0 = -\frac{\text{adj}(T)\ell}{\text{Re det } T},$$

and completing the Hermitian square gives

$$\min_{\zeta} Q(\zeta) = \frac{N}{\operatorname{Re} \det T}.$$

By (46), $N > 0$ when $\delta > 0$. Therefore $Q(\zeta) > 0$ for all ζ . Lemma 6.3 and the Schur slack (34) now give

$$\mathcal{P}(R) = Q(\zeta) + \lambda(c - \zeta^* G \zeta) > 0.$$

□

6.3 Closure of the Gram cone

Proof of Theorem 5.4. For $R \succeq 0$ and $t > 0$, the matrix $R + tI$ is positive definite, and hence $\mathcal{P}(R + tI) > 0$ by the preceding proposition. Since $\mathcal{P}$ is a polynomial in the real and imaginary parts of the entries, letting $t \downarrow 0$ gives $\mathcal{P}(R) \geq 0$. □

Proof of Theorem 5.2. If all coordinates of q are nonzero, the residual matrix (24) is positive semidefinite. Theorem 5.4 and (28) give (23).

For arbitrary q , choose phases p_i of modulus one, taking $p_i = q_i/|q_i|$ when $q_i \neq 0$ and $p_i = 1$ otherwise. Convexly interpolate the block matrix (22) with the rank-one correlation completion generated by $(p_1, p_2, p_3, 1)$. Its new last column is $q_i(t) = (1 - t)q_i + tp_i$, which is nonzero for $0 < t < 1$. The interpolated matrix remains positive semidefinite, so the already proved case applies. Letting $t \downarrow 0$ and using continuity proves the result. □

6.4 The resulting matrix certificate

The four-vector inequality has a useful matrix consequence. We record it after the direct proof so that it does not obscure the residual-vector argument that establishes it.

Lemma 6.6 (Ellipsoid-to-adjugate certificate). *Let $B, K \in M_m(\mathbb{C})$ with $B \succeq 0$ and $K = K^*$, and let $c \in \mathbb{R}$. If*

$$\begin{pmatrix} B & q \\ q^* & 1 \end{pmatrix} \succeq 0 \quad \implies \quad c + q^* K q \geq 0$$

for every $q \in \mathbb{C}^m$, then

$$(\det B)K + c \operatorname{adj}(B) \succeq 0. \tag{48}$$

Proof. If $\det B = 0$, the choice $q = 0$ gives $c \geq 0$ and $\operatorname{adj}(B) \succeq 0$, so (48) follows. If $\det B > 0$, then $B \succ 0$ and the block condition is $q^* B^{-1} q \leq 1$. Given $y \neq 0$, set

$$q = \frac{y}{\sqrt{y^* B^{-1} y}}.$$

The hypothesis, multiplied by $y^* B^{-1} y$, says $y^* (K + cB^{-1}) y \geq 0$. Hence $K + cB^{-1} \succeq 0$. Multiplication by $\det B > 0$ and the identity $\operatorname{adj}(B) = (\det B)B^{-1}$ prove the claim. □

Applied to (19)–(20), Lemma 6.6 gives the positive-semidefinite certificate

$$(\det B(u, v, w))K(u, v, w) + c_0(u, v, w) \operatorname{adj}(B(u, v, w)) \succeq 0. \tag{49}$$

7 Completion of the proof

No new inequality is needed in this section. We now follow the roadmap in reverse: the geometric polynomial proves the four-vector inequality, that inequality settles the two exceptional character rows, and the finite classification then recovers every subgroup and irreducible representation.

Proposition 7.1 (The two non-real A_4 rows). *For every 4×4 positive-semidefinite correlation matrix A ,*

$$d_{\chi_\omega}^{A_4}(A) \leq \text{per } A, \quad d_{\bar{\chi}_\omega}^{A_4}(A) \leq \text{per } A.$$

Proof. Write A in the block form (22) with $B = B(a, b, d)$ and $q = (c, e, f)^T$. Lemma 5.1 and Theorem 5.2 give the first inequality. An odd simultaneous relabeling preserves positivity and the permanent and interchanges the two non-real characters, giving the second. $\square$

Proof of Theorem 2.2. By Lemma 2.4, it is enough to consider correlation matrices. Lemma 3.1 and the transport Lemma 3.2 reduce an arbitrary subgroup and irreducible representation to one of the thirty-seven registered rows. Proposition 4.3 proves thirty-five of them and Proposition 7.1 proves the remaining two. Lemma 2.1 shows that the compared quantities are real. Finally, Lemma 2.3 returns from correlation matrices to the full positive-semidefinite cone. $\square$

A Character data

This appendix records the non-cyclic character data used in the finite row reduction. For S_4 , order the conjugacy classes as 1, (12), (12)(34), (123), (1234). Its irreducible character table is

	1	(12)	(12)(34)	(123)	(1234)
[4]	1	1	1	1	1
[1 ⁴]	1	-1	1	1	-1
[31]	3	1	-1	0	-1
[211]	3	-1	-1	0	1
[22]	2	0	2	-1	0

(50)

For A_4 , let C_+ and C_- be the two conjugacy classes of three-cycles. With ω as in (4), the table is

	1	(12)(34)	C_+	C_-
1	1	1	1	1
χ_ω	1	1	ω	ω^2
$\bar{\chi}_\omega$	1	1	ω^2	ω
χ_3	3	-1	0	0

(51)

For $D_8 = \langle r, s : r^4 = s^2 = 1, srs = r^{-1} \rangle$, the four linear characters take independent signs on r and s , while its two-dimensional character is 2, -2, 0, 0, 0 on

$$1, \quad r^2, \quad \{r, r^3\}, \quad \{s, r^2s\}, \quad \{rs, r^3s\}.$$

The standard character tables of C_m , C_2^2 , and S_3 supply the remaining rows listed in Section 4.

B Exact scalar and Gram certificates

This appendix records the non-obvious polynomial certificates used in Lemma 6.4. They contain no numerical approximation. Recall $\nu = X^2 + Y^2/3$.

B.1 Direct square completions

The coefficient B_2 has the exact decomposition

$$B_2 = 38 \left(\nu + \frac{39}{19}X - \frac{21}{19}Y + \frac{3}{2} \right)^2 + \frac{448}{19} \left(Y + \frac{3}{32}X - \frac{1083}{448} \right)^2 \\ + \frac{3483}{16} \left(X + \frac{2137}{2322} \right)^2 + \frac{131797}{5418} > 0.$$

Likewise

$$B_3 + 252 = 6 \left(\nu + \frac{5}{3}X - 2Y - \frac{1}{2} \right)^2 + 24 \left(Y - \frac{11}{12}X - \frac{15}{4} \right)^2 \\ + \frac{139}{6} \left(X - \frac{249}{139} \right)^2 + \frac{1851}{278} > 0.$$

Further elementary completions give

$$B_1 \geq \frac{112}{3}\nu \geq 0, \quad B_4 \geq \frac{132}{7} > 0.$$

B.2 The conditional middle-coefficient certificate

Set

$$\Sigma = 4B_2B_4 + 252B_3.$$

An exact Gram decomposition is

$$\Sigma = 2128\nu \left(\nu + \frac{387}{133}X - \frac{261}{133}Y + \frac{77}{20} \right)^2 + \frac{1}{59850}m^T Q m, \quad (52)$$

where $m = (1, Y, Y^2, X, XY, X^2)^T$ and

$$Q = \begin{pmatrix} 8739057600 & -4231634400 & 478800000 & 6265576800 & -1795500000 & 47880000 \\ -4231634400 & 2519799114 & -385003080 & -2526627600 & 807975000 & 239400000 \\ 478800000 & -385003080 & 79782560 & 281630160 & -81252000 & -35910000 \\ 6265576800 & -2526627600 & 281630160 & 10931106942 & -2359670040 & 2303554680 \\ -1795500000 & 807975000 & -81252000 & -2359670040 & 685371360 & -243756000 \\ 47880000 & 239400000 & -35910000 & 2303554680 & -243756000 & 1122611040 \end{pmatrix}.$$

Exact rational LDL^T elimination has positive pivots

$$\begin{aligned} & 8739057600, \quad \frac{79557301866}{169}, \quad \frac{1112696849800960}{299087601}, \\ & \frac{2624099206828303859949}{869294413907}, \quad \frac{2531308970961028469920080}{115380521779374043}, \\ & \frac{1406694633431943161674826341030}{10547120712337618624667}. \end{aligned}$$

Thus $Q \succ 0$ and $m^T Q m > 0$ because the first coordinate of m is one. Equation (52) proves $\Sigma > 0$.

If $B_3 \geq 0$, then $B_2 + aB_3 + a^2B_4 > 0$ is immediate. If $B_3 < 0$, the identity

$$4B_2B_4 - B_3^2 = (4B_2B_4 + 252B_3) + (-B_3)(B_3 + 252)$$

shows that the discriminant of the quadratic $B_2 + aB_3 + a^2B_4$ is negative. Hence that quadratic is positive for every $a > 0$. This completes the proof that $N_1 > 0$.

C Character-row audit

For convenience, the logical dispatch is summarized below.

subgroup types	rows	analytic certificate
$1, C_2, C_3, V_4, S_3$	19 rows	principal 2×2 and 3×3 minor inequalities and principal-permanent contraction
C_4	4 rows	opposite-edge AM–GM plus the two block contractions
D_8	5 rows	symmetric and exterior $2 + 2$ block contractions
S_4	5 rows	eigenvalue moment identity, cycle bounds, and per $A \geq 1$
A_4	trivial and 3-dimensional	averages of already proved S_4 rows
A_4	$\chi_\omega, \bar{\chi}_\omega$	residual-vector geometric polynomial and Theorem 5.2

The row count in the first entry is $1 + 2 + 2 + 3 + 4 + 4 + 3 = 19$; adding $4 + 5 + 5 + 4$ gives 37. The A_4 line contains four rows in total, two routine and two geometric.

D Correspondence with the Lean formalization

The proof has been formalized in Lean 4.19.0 with Mathlib 4.19.0. The main mathematical correspondences are:

manuscript component	Lean modules
definitions and normalization	Basic, DiagonalScaling, CorrelationLift
subgroups and character completeness	SubgroupRegistry, CharacterTables, realization and completeness registries
routine analytic rows	LowOrderRegistry, C4TableBridge, D8TableBridge, S4TableBridge
A_4 expansion and certificate	A4Certificate, A4Gap, A4GeometricBridge
residual normalization	A4GeometricNormalization, A4GeometricFourVector
rank-one boundary	A4GeometricRankOne, A4GeometricCollinear
Hessian and minimum	A4GeometricHessian, A4GeometricQuadratic, A4GeometricMinimum
exact polynomial certificates	A4GeometricGeneral, A4GeometricGram, A4GeometricAssembly
PSD closure and final theorem	A4GeometricPSD, FinalAssembly, NormalizedPermanentalDominance

The public formal endpoints are

`PermanentalDominance.N4.normalized_permanental_dominance_n4`
`PermanentalDominance.N4.normalized_permanental_dominance_n4_real`

and the central geometric endpoints are

`A4GeometricPSD.geometricP_nonneg_of_posSemidef`
`A4GeometricFourVector.fourVectorProperty`
`A4GeometricBridge.certificate_posSemidef_geometric`